Aula 11: Modelos VAR: Funções de Resposta ao Impulso

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Causalidade de Granger

- Como saber se uma variável é capaz de prever outra? Se sim, em quais condições?
- **Granger-causalidade**: indica se movimentos na variável y antecedem os movimentos da variável z.
- Por que isso é útil?
- **Nota**: Jamais confundir com efeito causal de uma política ou intervenção!
- O teste de Granger é bem simples: um teste F conjunto para os coeficientes de interesse, assumindo SEMPRE que eles multipliquem variáveis estacionárias:

$$z_t = \phi_{20} + \sum_{i=1}^{p} \phi_{i,21} y_{t-i} + \sum_{i=1}^{p} \phi_{i,22} z_{t-i} + e_{2t}$$

$$\begin{cases} H_0 : \phi_{1,21} = \phi_{2,21} = \ldots = \phi_{p,21} = 0 \\ H_1 : \phi_{i,21} \neq 0, \; i = 1, 2, \ldots, p \end{cases}$$

- A estatística de teste é dada por:

$$S_1 = \frac{(e_r^2 - e_u^2)/p}{e_u^2/(T - 2p - 1)} \sim F(p, T - 2p - 1)$$

  em que r representa o modelo restrito e u representa o modelo nao-restrito. Se S_1 for maior do que o nível de significância, rejeita-se a nula de que y não Granger-causa z.

- Causalidade de Granger não é o mesmo que teste de exogeneidade. Granger mede precedência temporal. Exogeneirade requer que as variáveis não sejam afetadas contemporaneamente e um modelo VAR em forma reduzida não permite testar isso.
- Exemplo de teste de causalidade de Granger no R:

```r
library(rbcb)
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
ipca_table = get_series(433, start_date = '2000-01-01')
ipcats = ts( ipca_table$`433`, start = c( 2000, 1 ), frequency = 12 )      # IPCA

selic_table <- get_series(4189, start_date = '2000-01-01')
selicts <- ts(selic_table$`4189`, start = c(2000, 1), frequency = 12)      # Selic
```

```r
library(lmtest)
```

```
## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
dbvar <- ts.intersect(ipcats, selicts)

# Executar o teste de causalidade de Granger
result1 <- grangertest(ipcats ~ selicts, order = 10, data = dbvar)

result2 <- grangertest(selicts ~ ipcats, order = 10, data = dbvar)

# Visualizar os resultados
print(result1)
```

```
## Granger causality test
##
## Model 1: ipcats ~ Lags(ipcats, 1:10) + Lags(selicts, 1:10)
## Model 2: ipcats ~ Lags(ipcats, 1:10)
##   Res.Df  Df      F   Pr(>F)
## 1    278
## 2    288 -10 2.6546 0.004143 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
print(result2)
```

```
## Granger causality test
##
## Model 1: selicts ~ Lags(selicts, 1:10) + Lags(ipcats, 1:10)
## Model 2: selicts ~ Lags(selicts, 1:10)
##   Res.Df  Df      F  Pr(>F)
## 1    278
## 2    288 -10 1.6253 0.09904 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

VAR Estrutural

- A Decomposição de Cholesky permite supor parâmetros estruturais do VAR a partir da diagonalização da matriz de covariância dos resíduos.
- Entretanto, nem sempre teremos um número suficiente de restrições para montar esta matriz diagonal!
- Por exemplo, nem sempre podemos assumir efeito nulo das relações econômicas que buscamos estimar.
- Vamos agora introduzir um modelo mais genérico, que permite testar diferentes restrições. Partindo de um VAR(1) com as seguintes equações:

$$AX_t = B_0 + B_1 X_{t-1} + B\varepsilon_t$$

Quando se estima a forma reduzida, obtêm-se os resíduos $\hat{e}_t$ tal que:

$$B\hat{\varepsilon}_t = A\hat{e}_t$$

- O problema é restringir o sistema de forma a recuperar ϵ_t segundo a hipótese de que cada resíduo da forma estrutural é independente.
- Para começar, usa-se a matriz de covariância dos erros de forma reduzida:

$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \dots & \sigma_{nn} \end{pmatrix}$$

- Em que cada elemento é estimado por:

$$\hat{\sigma}_{ij} = \frac{1}{T}\sum_{t=1}^{T} \hat{e}_{it}\hat{e}_{jt}$$

- Após calcular a matriz de variância-covariância dos resíduos, podemos usar o seguinte resultado para recuperar os parâmetros estruturais:

$$BB' = \text{Var}(Ae_t) = A\text{Var}(e_t)A'$$

- A matriz de covariância dos choques estruturais BB' tem n variâncias desconhecidas. Assumido que a matriz A tem diagonal unitária (normalização), a matriz A tem $(n^2 - n)$ elementos desconhecidos. Então são n^2 elementos desconhecidos.
- Conhecemos a matriz $\Sigma = Var(e_t)$ que tem $\frac{n(n+1)}{2}$ elementos conhecidos.
- Logo, são necessárias $\frac{n(n+1)}{2}$ restrições para identificação exata do modelo estrutural!
- Exemplo 6.13 (BUENO): Monte o sistema de equações e imponha a restrição de Cholesky para resolvê-lo:

$$\begin{pmatrix} \sigma^2_{\epsilon_1} & 0 \\ 0 & \sigma^2_{\epsilon_2} \end{pmatrix} = \begin{pmatrix} 1 & a_{12} \\ a_{21} & 1 \end{pmatrix} \begin{pmatrix} 0.5 & 0.4 \\ 0.4 & 0.6 \end{pmatrix} \begin{pmatrix} 1 & a_{21} \\ a_{12} & 1 \end{pmatrix}$$

Exemplo no R: identificando modelos VAR:

passo 1: montando a base de dados

```r
library(rbcb)
library(tseries)


ipca_table = get_series(433, start_date = '2000-01-01')
```

```
## Skipping download - using cached version
```

```r
ipcats = ts( ipca_table$`433`, start = c( 2000, 1 ), frequency = 12 )      # IPCA

ibc_table <- get_series(24364, start_date = '2003-01-01')
idxibcts <- ts(ibc_table$`24364`, start = c(2003, 1), frequency = 12)      # IBC-BR
ibcts <- 100*diff(log(idxibcts))

selic_table <- get_series(4189, start_date = '2000-01-01')
```

```
## Skipping download - using cached version
```

```r
selicts <- ts(selic_table$`4189`, start = c(2000, 1), frequency = 12)     # Selic


dbvar <- ts.intersect(ibcts, ipcats, selicts)
dbvar <- ts(dbvar, start = start(dbvar), frequency = 12)
```

Passo 2: selecionando a ordem do VAR:

```r
library(vars)
```

```
## Loading required package: MASS

## Loading required package: strucchange

## Loading required package: sandwich

## Loading required package: urca
```

```r
# Selecionando a ordem do modelo VAR
var_order <- VARselect(dbvar, lag.max = 12, type = "const")
print(var_order)
```

```
## $selection
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      5      3      3      5
##
## $criteria
##                  1            2            3            4            5
## AIC(n) -4.44421805 -5.419349450 -5.544040577 -5.526019493 -5.594454674
## HQ(n)  -4.37796068 -5.303399052 -5.378397152 -5.310683040 -5.329425194
## SC(n)  -4.27942293 -5.130957985 -5.132052771 -4.990435344 -4.935274184
## FPE(n)  0.01174637  0.004430203  0.003911144  0.003982816  0.003720184
##                  6           7            8            9           10
## AIC(n) -5.572485463 -5.55062358 -5.516330598 -5.533075299 -5.524648699
## HQ(n)  -5.257762955 -5.18620804 -5.102222035 -5.069273708 -5.011154080
## SC(n)  -4.789708631 -4.64425041 -4.486361082 -4.379509441 -4.247486499
## FPE(n)  0.003804033  0.00388982  0.004027853  0.003963886  0.004001086
##                 11           12
## AIC(n) -5.506787388 -5.476241882
## HQ(n)  -4.943599742 -4.863361208
## SC(n)  -4.106028847 -3.951886999
## FPE(n)  0.004077727  0.004209807
```

Passo 3: estimando o VAR

```r
ordem = 3;

# Estimando o modelo VAR
var_model <- VAR(dbvar, p = ordem, type = "const")
summary(var_model)
```

```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: ibcts, ipcats, selicts
## Deterministic variables: const
## Sample size: 268
```

```
## Log Likelihood: -404.541
## Roots of the characteristic polynomial:
## 0.9204 0.9204 0.5489 0.5489 0.5031 0.5031 0.4302 0.4302 0.1622
## Call:
## VAR(y = dbvar, p = ordem, type = "const")
##
##
## Estimation results for equation ibcts:
## ======================================
## ibcts = ibcts.l1 + ipcats.l1 + selicts.l1 + ibcts.l2 + ipcats.l2 + selicts.l2 + ibcts.l3 + ipcats.l3
##
##            Estimate Std. Error t value Pr(>|t|)
## ibcts.l1    0.21098    0.06104   3.456 0.000640 ***
## ipcats.l1  -0.12400    0.25776  -0.481 0.630888
## selicts.l1  0.08656    0.30339   0.285 0.775645
## ibcts.l2   -0.12221    0.06246  -1.957 0.051477 .
## ipcats.l2   0.50023    0.28829   1.735 0.083906 .
## selicts.l2 -0.30929    0.55696  -0.555 0.579156
## ibcts.l3   -0.05510    0.06042  -0.912 0.362575
## ipcats.l3  -0.92915    0.26198  -3.547 0.000463 ***
## selicts.l3  0.22892    0.28748   0.796 0.426590
## const       0.33942    0.22051   1.539 0.124969
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 1.123 on 258 degrees of freedom
## Multiple R-Squared: 0.1159,  Adjusted R-squared: 0.08503
## F-statistic: 3.757 on 9 and 258 DF,  p-value: 0.0001839
##
##
## Estimation results for equation ipcats:
## =======================================
## ipcats = ibcts.l1 + ipcats.l1 + selicts.l1 + ibcts.l2 + ipcats.l2 + selicts.l2 + ibcts.l3 + ipcats.l3
##
##             Estimate Std. Error t value Pr(>|t|)
## ibcts.l1    0.031302   0.014892   2.102   0.0365 *
## ipcats.l1   0.502408   0.062886   7.989 4.53e-14 ***
## selicts.l1  0.034758   0.074018   0.470   0.6390
## ibcts.l2   -0.011437   0.015238  -0.751   0.4536
## ipcats.l2   0.048029   0.070334   0.683   0.4953
## selicts.l2  0.008864   0.135883   0.065   0.9480
## ibcts.l3    0.015277   0.014740   1.036   0.3009
## ipcats.l3  -0.069791   0.063915  -1.092   0.2759
## selicts.l3 -0.046561   0.070136  -0.664   0.5074
## const       0.266664   0.053799   4.957 1.30e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.274 on 258 degrees of freedom
## Multiple R-Squared: 0.3202,  Adjusted R-squared: 0.2965
## F-statistic:  13.5 on 9 and 258 DF,  p-value: < 2.2e-16
##
```

```
##
## Estimation results for equation selicts:
## ========================================
## selicts = ibcts.l1 + ipcats.l1 + selicts.l1 + ibcts.l2 + ipcats.l2 + selicts.l2 + ibcts.l3 + ipcats.l
##
##             Estimate Std. Error t value Pr(>|t|)
## ibcts.l1    0.026827   0.012111   2.215 0.027622 *
## ipcats.l1   0.023595   0.051140   0.461 0.644918
## selicts.l1  1.582214   0.060193  26.286  < 2e-16 ***
## ibcts.l2    0.006247   0.012392   0.504 0.614597
## ipcats.l2   0.163053   0.057197   2.851 0.004715 **
## selicts.l2 -0.400806   0.110503  -3.627 0.000345 ***
## ibcts.l3    0.026823   0.011987   2.238 0.026091 *
## ipcats.l3   0.019791   0.051977   0.381 0.703687
## selicts.l3 -0.196544   0.057036  -3.446 0.000664 ***
## const       0.056891   0.043750   1.300 0.194640
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.2228 on 258 degrees of freedom
## Multiple R-Squared: 0.9975,  Adjusted R-squared: 0.9974
## F-statistic: 1.153e+04 on 9 and 258 DF,  p-value: < 2.2e-16
##
##
##
## Covariance matrix of residuals:
##           ibcts   ipcats  selicts
## ibcts   1.26150 0.031020 0.015525
## ipcats  0.03102 0.075087 0.005485
## selicts 0.01552 0.005485 0.049657
##
## Correlation matrix of residuals:
##           ibcts  ipcats selicts
## ibcts   1.00000 0.10079 0.06203
## ipcats  0.10079 1.00000 0.08983
## selicts 0.06203 0.08983 1.00000
```

Passo 4: identificação do VAR (os resíduos do VAR são $\hat{e}_t$)

$$A\hat{e}_t = B\hat{\varepsilon}_t$$

- Por Cholesky (depende da ordem das variáveis):

```r
library(svars)
```

```
## Registered S3 method overwritten by 'svars':
##   method           from
##   stability.varest vars
```

```r
svar_chol <- id.chol(var_model)

svar_chol
```

```
##
## Estimated B Matrix (unique decomposition of the covariance matrix):
```

```
##                [,1]       [,2]      [,3]
## ibcts   1.12316707 0.00000000 0.0000000
## ipcats  0.02761832 0.27262480 0.0000000
## selicts 0.01382245 0.01871924 0.2216212
```

```r
# Alternativa usando função SVAR:

b_chol <- svar_chol$B
diag(b_chol) <- NA
b_chol[2,1] <- NA
b_chol[3,1:2] <- NA

svar_chol2 <- SVAR(var_model, Amat = NULL, Bmat = b_chol, estmethod = "scoring")
```

```
## Warning in SVAR(var_model, Amat = NULL, Bmat = b_chol, estmethod = "scoring"):
## The B-model is just identified. No test possible.
```

```r
svar_chol2
```

```
##
## SVAR Estimation Results:
## ========================
##
##
## Estimated B matrix:
##           ibcts  ipcats selicts
## ibcts   1.12317 0.00000  0.0000
## ipcats  0.02762 0.27262  0.0000
## selicts 0.01382 0.01872  0.2216
```

- Por restrições de curto prazo

```r
A_mat <- NULL

B_mat <- matrix(NA, 3, 3)
diag(B_mat) <- 1

B_mat[1,2] <- 0 # Exemplo: (2:IPCA não afeta  1:ATIVIDADE contemporaneamente)
B_mat[1,3] <- 0 # Exemplo: (3:SELIC não afeta 1:ATIVIDADE contemporaneamente)
B_mat[2,3] <- 0 # Exemplo: (3:SELIC não afeta 2:IPCA contemporaneamente)

svar_ab <- SVAR(var_model, Amat = A_mat, Bmat = B_mat, estmethod = "scoring")
```

```
## Warning in SVAR(var_model, Amat = A_mat, Bmat = B_mat, estmethod = "scoring"):
## Convergence not achieved after 100 iterations. Convergence value:
## 1.42918301324069e-06 .
```

```r
svar_ab
```

```
##
## SVAR Estimation Results:
## ========================
##
##
## Estimated B matrix:
##           ibcts  ipcats selicts
## ibcts   1.00000 0.00000       0
## ipcats  0.02459 1.00000       0
```

```
## selicts 0.01231 0.06868       1
```

Compare as IRFs dos dois modelos com estratégias diferentes de identificação:

```r
# IRFs para ambos os modelos
irf_chol <- irf(svar_chol2, impulse = NULL, response = NULL, n.ahead = 24, boot = TRUE, ortho = TRUE)
```

```
## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = NULL, Bmat =
## b_chol): The B-model is just identified. No test possible.
## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = NULL, Bmat =
## b_chol): The B-model is just identified. No test possible.
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## b_chol): The B-model is just identified. No test possible.
## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = NULL, Bmat =
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## b_chol): The B-model is just identified. No test possible.
```

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## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = NULL, Bmat =
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## b_chol): The B-model is just identified. No test possible.
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## b_chol): The B-model is just identified. No test possible.
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## b_chol): The B-model is just identified. No test possible.
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## b_chol): The B-model is just identified. No test possible.
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## b_chol): The B-model is just identified. No test possible.
```

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## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = NULL, Bmat =
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## b_chol): The B-model is just identified. No test possible.
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## b_chol): The B-model is just identified. No test possible.
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## b_chol): The B-model is just identified. No test possible.
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## b_chol): The B-model is just identified. No test possible.
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## b_chol): The B-model is just identified. No test possible.
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## b_chol): The B-model is just identified. No test possible.
## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = NULL, Bmat =
## b_chol): The B-model is just identified. No test possible.
```
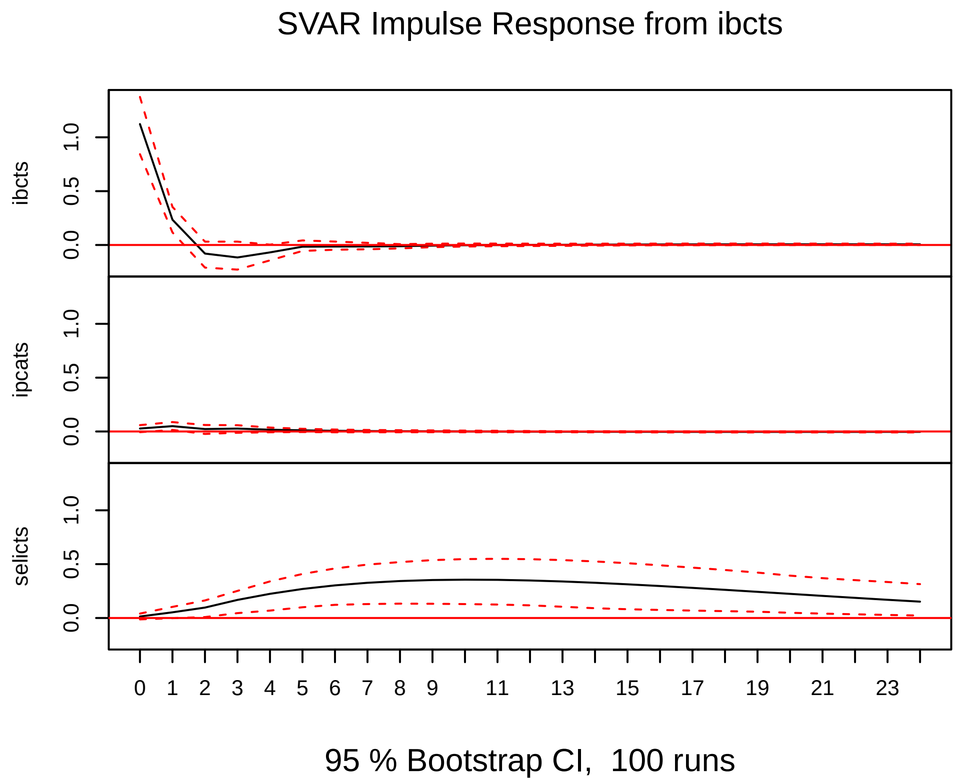
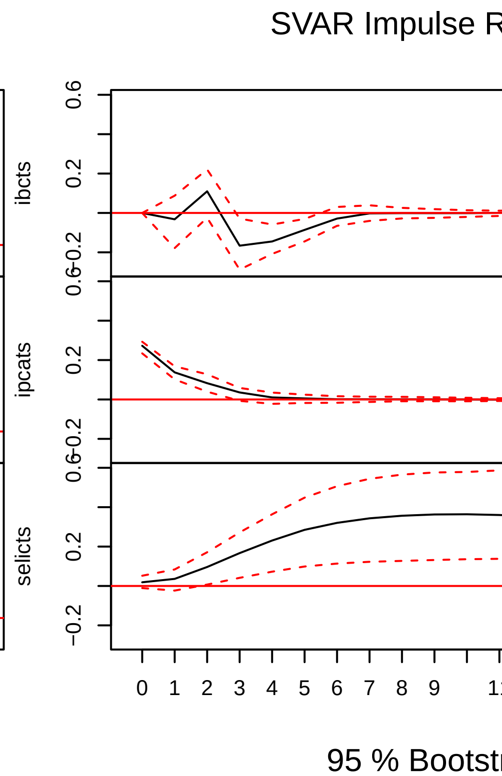
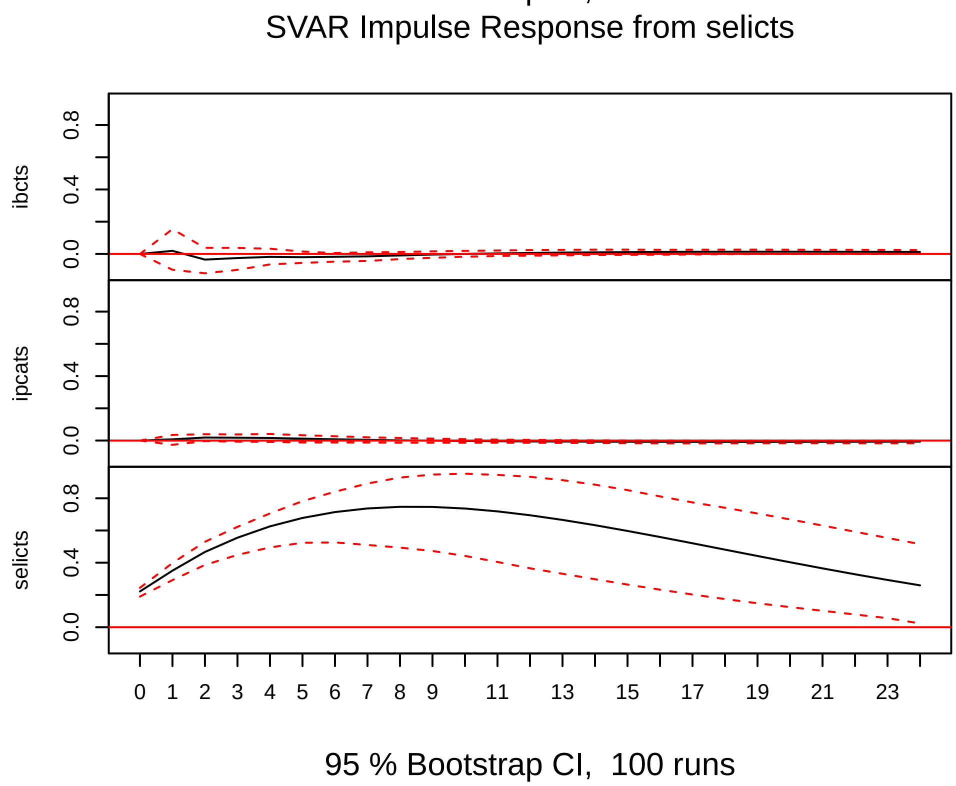

```r
plot(irf_chol)
```








```r
irf_ab <- irf(svar_ab, impulse = NULL, response = NULL, n.ahead = 24, boot = TRUE, ortho = TRUE)
```

```
## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.08490580026854e-06 .
```

```
## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.66796581799536e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.52097612121649e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 9.77644289093527e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.35585093660739e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.2062458074058e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.82668387412244e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.10400401907382e-05 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.99046852308971e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.51570166816956e-05 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.31786082990804e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 6.0132208778152e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 5.30072150789485e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 4.93315124230204e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 4.51124640091227e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.71230748774654e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
```

```
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 4.41018890684486e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 4.04132645905747e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.33390420034774e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.03464965674205e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 4.20843301203108e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.70547862275816e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.03907741287948e-05 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.69230247181007e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 4.18823738482232e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.86432636850586e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.85075585606986e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 6.45956902550759e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.42799575483799e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.2717587639377e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.20968224028919e-05 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
```

```
## 4.43874009811718e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 7.23448245532055e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 6.79637842672387e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 9.29666739389601e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.15102362566177e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 7.83795710404545e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 7.10172640069895e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 5.14802924835023e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 6.28919063434125e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.20362305294974e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 8.24872268198606e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.46865733810508e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.24720038897908e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.55562057580216e-05 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 5.14957518127357e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.69879439728265e-05 .
```

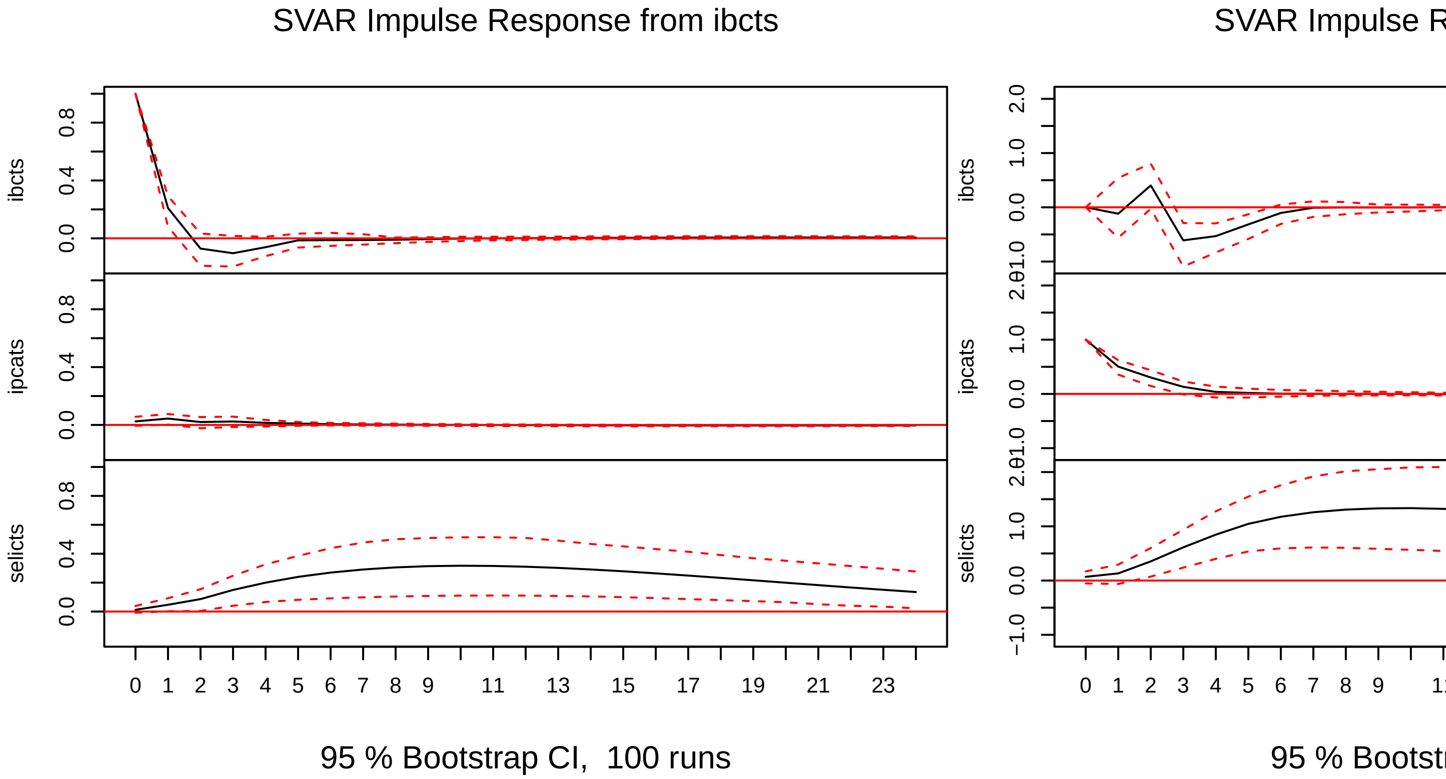
```
## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 9.49478554612693e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 6.50933953377786e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.36095158825944e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.01320325944609e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 7.8499210145605e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 7.00542017298411e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.28809576457395e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.92346702154733e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.54073323022286e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 7.46396034641272e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.01879251640513e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 9.16243184212462e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 8.61192014880996e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.49447703676398e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.93732697700944e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
```

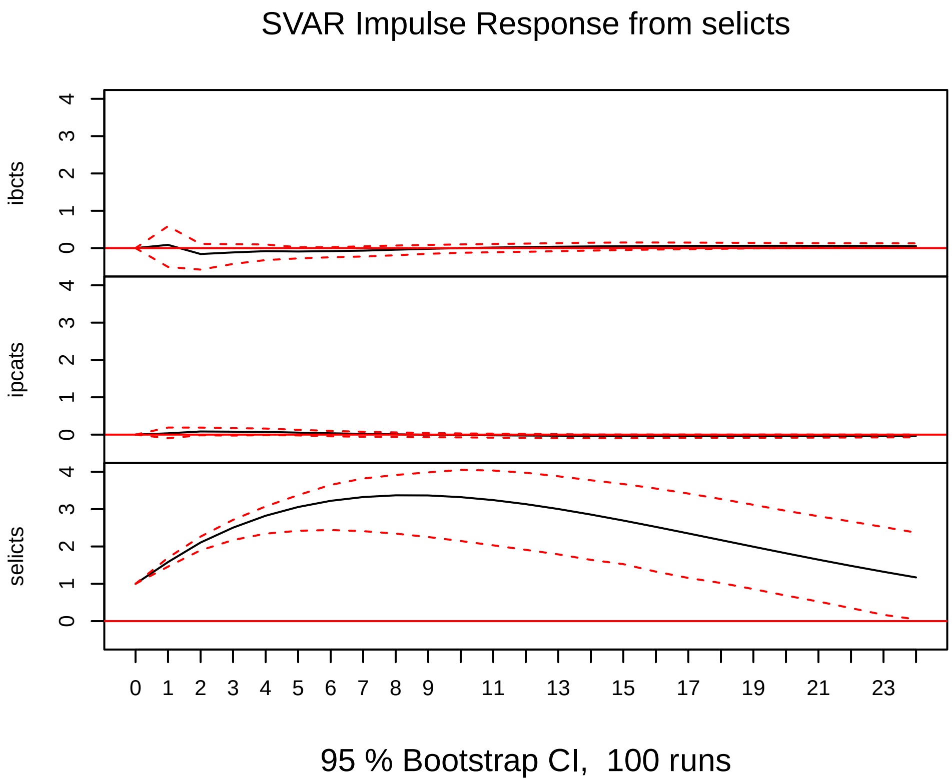
```
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.51111847145158e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.1324542426111e-05 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.69253391085955e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.79627118943585e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 4.97665394866287e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 5.3137659770941e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 5.53431578814712e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.03729088073257e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.19319750108149e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.82778941604334e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 5.44610754771302e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.67001006384782e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.87190986752905e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 4.95283425392193e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 6.89707260756759e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
```

```
## 1.49934483475861e-05 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.96630757426375e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 9.98799830037589e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 4.39564663903824e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.32261907325598e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.18506746210712e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.36119456609074e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.92007428467911e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 4.52269124305571e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.96774040359116e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 8.81471942396506e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.24428620149153e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 3.5244321837935e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.43730488282034e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.42372833392135e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.85775565008858e-06 .
```

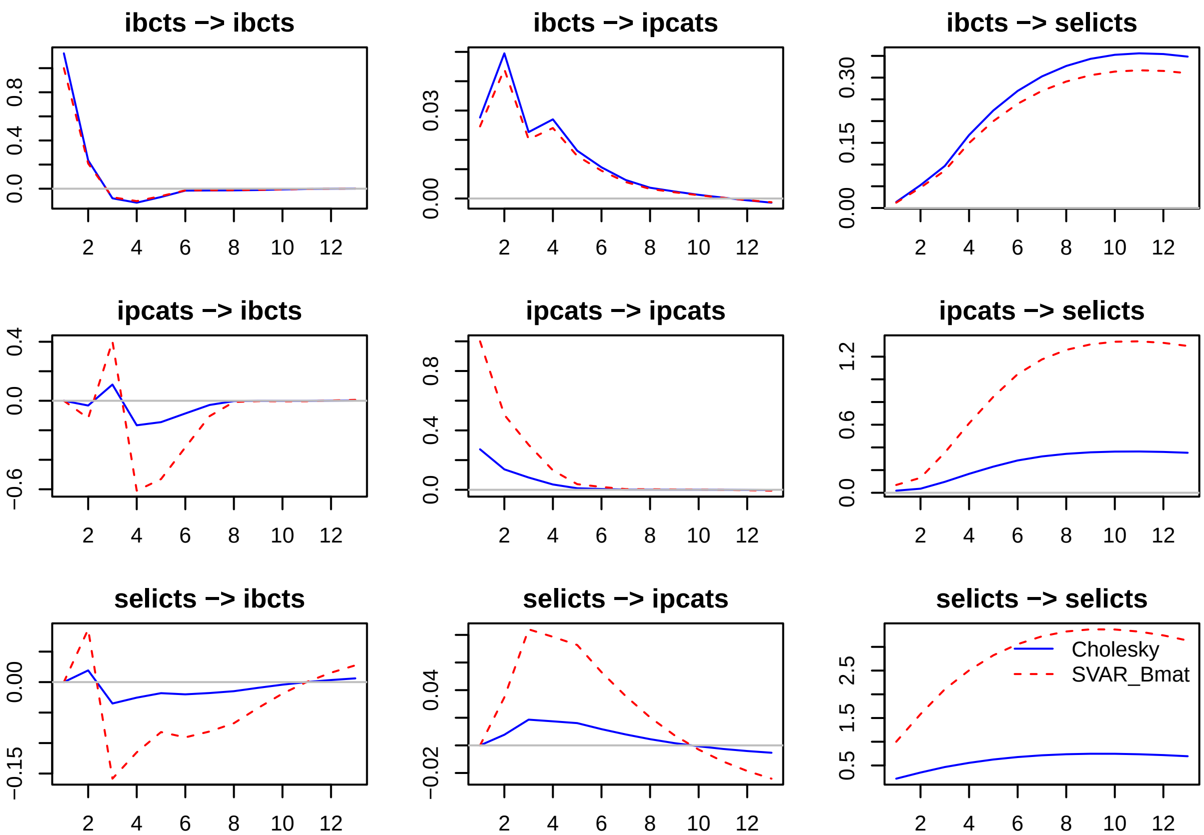
```
## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.01237139767812e-06 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 8.40275184085804e-07 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 1.37166979998788e-05 .

## Warning in SVAR(x = varboot, estmethod = "scoring", Amat = A_mat, Bmat =
## B_mat): Convergence not achieved after 100 iterations. Convergence value:
## 2.02801575666101e-06 .
```

```r
plot(irf_ab)
```






```r
# Função para plotar IRFs comparando os dois modelos
plot_irf_comparison <- function(svar1, svar2, legend, n.ahead = 24) {
  vars <- colnames(svar1$var$y)
  par(mfrow = c(length(vars), length(vars)), mar = c(3, 3, 2, 1))
  for (impulse in vars) {
    for (response in vars) {
      irf1 <- irf(svar1, impulse = impulse, response = response, n.ahead = n.ahead, boot = FALSE, ortho
      irf2 <- irf(svar2, impulse = impulse, response = response, n.ahead = n.ahead, boot = FALSE, ortho
      plot(irf1$irf[[1]], type = "l", col = "blue", ylim = range(c(irf1$irf[[1]], irf2$irf[[1]])),
           main = paste(impulse, "→", response), ylab = "", xlab = "")
      lines(irf2$irf[[1]], col = "red", lty = 2)
      abline(h = 0, col = "gray")
    }
  }
  legend("topright", legend = legend, col = c("blue", "red"), lty = c(1, 2), bty = "n")
}

# Gerar os gráficos
plot_irf_comparison(svar_chol2, svar_ab, c("Cholesky", "SVAR_Bmat"), n.ahead = 12)
```



O problema da identificação dos modelos SVAR

- VAR tem uma especificação flexível para capturar a dinâmica dos dados. Todavia, os resíduos não permitem analisar os efeitos de cada equação/choque.
- Não conseguimos identificar/mapear todos os parâmetros de um modelo estrutural sem impor restrições adicionais.
- Diversas abordagens/estratégias têm sido desenvolvidas e são bastante utilizadas para análise multivariada e identificação de choques estruturais.
    1. Restrições de curto prazo por meio da identificação recursiva (decomposição de Cholesky) (Sims 1980; Blanchard 1989; Christiano, Eichenbaum & Evans 1999; Bjørnland 2009)
    2. Restrições de curto prazo com ajuda da literatura (teoria + evidência empírica)
    3. Restrições de longo prazo (Blanchard and Quah 1989; Gali 1999)
    4. Restrições de sinais (Uhlig 2005)
    5. Instrumentos externos
    6. Combinar restrições de sinais com instrumentos externos
    7. Narrativas

Blanchard & Quah (1989)

Seja X_t um vetor $n \times 1$ com n variáveis endógenas, tal que:

$$AX_t = B_0 + \sum_{i=1}^{p} B_i X_{t-i} + B\varepsilon_t$$

onde:

- A é uma matriz $n \times n$ de restrições contemporâneas,
- B_0 é um vetor $n \times 1$ de constantes,
- B_i é a matriz $n \times n$ da i-ésima defasagem,
- $\varepsilon_t \sim i.i.d.(0, I_n)$ é um vetor $n \times 1$ de termos de erro,

O modelo VAR equivalente em forma reduzida tem ordem p (Bueno, 2012):

$$X_t = \Phi_0 + \sum_{i=1}^{p} \Phi_i X_{t-i} + \epsilon_t$$

Se os autovalores do polinômio $I - \sum_{i=1}^{p} \Phi_i L^i$ estiverem fora do círculo unitário, podemos utilizar a **decomposição de Wold** e reescrever o modelo como um VMA(∞):

$$X_t = \mu + \sum_{j=0}^{\infty} \Phi_1^j \epsilon_{t-j}$$

Definindo $C_j = \Phi_1^j B$, temos:

$$X_t = \mu + \sum_{j=0}^{\infty} C_j \varepsilon_{t-j}$$

Agora vamos desenvolver essa equação matricial para o caso em que $n = 2$:

$$\begin{bmatrix} y_t \\ z_t \end{bmatrix} = \begin{bmatrix} \mu_y \\ \mu_z \end{bmatrix} + \sum_{j=0}^{\infty} \begin{bmatrix} c_{11,j} & c_{12,j} \\ c_{21,j} & c_{22,j} \end{bmatrix} \begin{bmatrix} \varepsilon_{y,t-j} \\ \varepsilon_{z,t-j} \end{bmatrix}$$

A derivada parcial de y_t em relação a $\varepsilon_{y,t}$ é:

$$\frac{\partial y_t}{\partial \varepsilon_{y,t}} = c_{11,0}$$

Para que os choques em y_t não tenham efeito de longo prazo, é necessário impor:

$$\sum_{j=0}^{\infty} c_{11,j} = 0$$

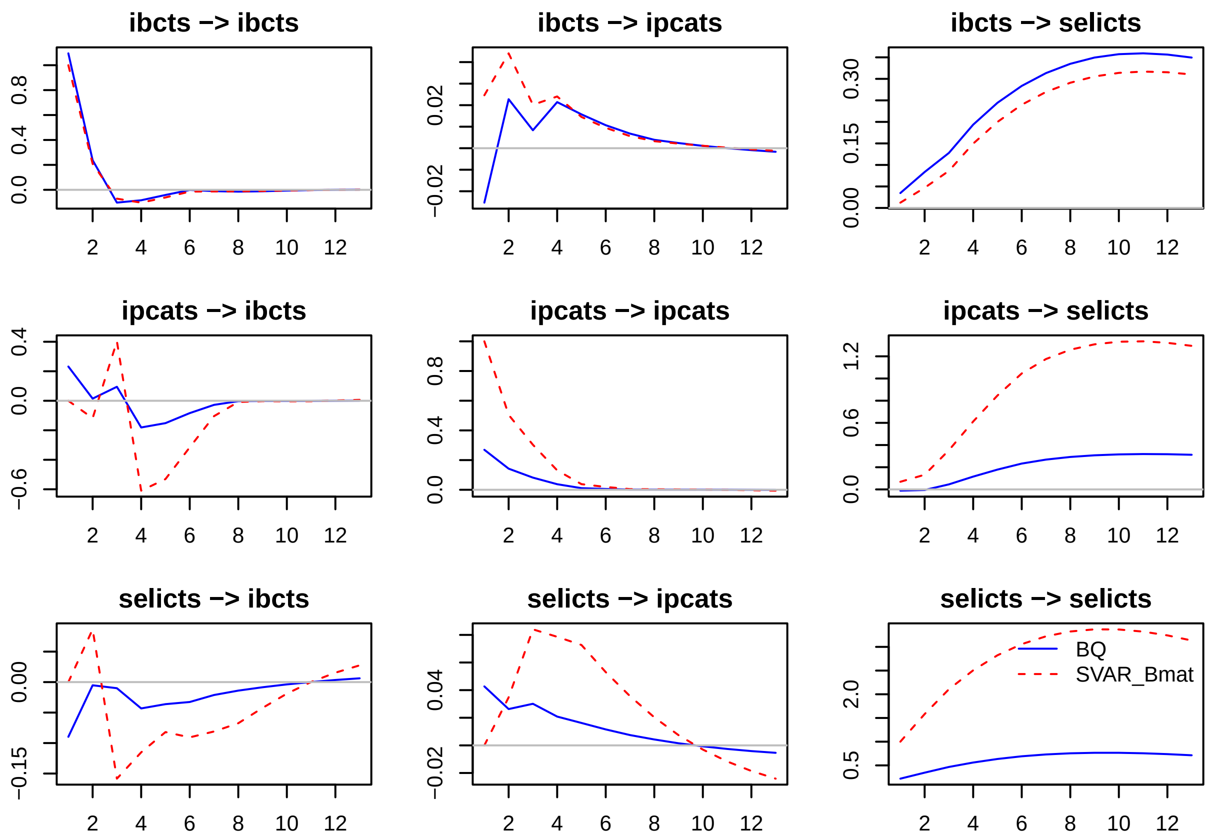
- **Blanchard & Quah (1989): o que gera flutuações na taxa de crescimento do PIB** (Δy_t) e na taxa de desemprego (U_t)?
  - Dois tipos de choques:
    - choques com efeitos permenantes no produto
    - choques com efeitos transitórios

- Blanchard & Quah interpretam os choques com efeitos permanentes no produto como choques de oferta (ε_t^{supply}) e os choques com efeitos transitórios no produto como choques de demanda (ε_t^{demand}). Ou seja, no longo prazo, apenas os choques de oferta afetam o PIB.
- Podemos impor essa condição de longo prazo para identificar o VAR!

```r
# 4. Estimar SVAR com identificação de longo prazo (Blanchard-Quah)

restricoes <- matrix(NA, 3, 3)
restricoes[2, 1] <- 0
restricoes[2, 3] <- 0

var_model_lr <- BQ(var_model)

var_model_lr
```

```
##
## SVAR Estimation Results:
## ========================
##
##
## Estimated contemporaneous impact matrix:
##            ibcts   ipcats  selicts
## ibcts    1.09524  0.23216 -0.08977
## ipcats  -0.02530  0.26949  0.04270
## selicts  0.03458 -0.01124  0.21985
##
## Estimated identified long run impact matrix:
##             ibcts ipcats selicts
## ibcts    1.181591 0.0000    0.00
## ipcats  -0.007575 0.4856    0.00
## selicts  6.856908 5.8807   14.53
```

```r
# Gerar os gráficos
plot_irf_comparison(var_model_lr, svar_ab, c("BQ","SVAR_Bmat"),n.ahead = 12)
```



- Outros exemplos de restrições de longo prazo, derivadas da teoria econômica, que podem ser utilizadas para identificar o VAR:
  - Neutralidade da moeda
  - Curva de Phillips vertical (choques de preços não afetam a taxa de desemprego no longo prazo)
  - Efeito Fisher (choques de preços não afetam a taxa de juros real)